DTC P0A0F:00 [SKYACTIV-D 1.5]

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DTC P0A0F: 00	**Engine failed to restart**
DETECTION CONDITION	• Any of following conditions occurs: — When the engine should restart after it is stopped by the i-stop control, it does not start even though it is cranked for **3 s or more**. — When the engine should restart after it is stopped by the i-stop control, it does not cranked. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Related parts malfunction — Starter relay malfunction — Battery malfunction — Wiring harness malfunction between starter relay and PCM, or poor connector contact — Starter malfunction • Common rail system malfunction • Piston stop position malfunction during idling stop — CKP sensor signal malfunction — Intake air system related malfunction (air suction, vacuum hose breakage) — HP-EGR control malfunction — LP-EGR control malfunction • Mechanical (engine) malfunction — Improper valve timing • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P1260:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P1260:00 [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT FOR A MALFUNCTION RELATED INABILITY TO START ENGINE** • Start the engine. • Does the engine start normally?	Yes	Go to the next step.
		No	Perform the symptom troubleshooting “NO.9 WILL NOT CRANK” or “NO.10 HARD TO START/LONG CRANK/ ERRATIC START/ERRATIC CRANK”. (See NO.9 WILL NOT CRANK [SKYACTIV-D 1.5].) (See NO.10 HARD TO START/LONG CRANK/ERRATIC START/ERRATIC CRANK [SKYACTIV-D 1.5].)
4	**INSPECT COMMON RAIL SYSTEM OPERATION** • Inspect the common rail operation. (See COMMON RAIL INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**DETERMINE IF MALFUNCTION IS CAUSED BY ROUGH IDLING OR A PISTON-STOP POSITION CONTROL MALFUNCTION** • Start the engine and warm it up completely. • Is the idle speed stabilized?	Yes	Go to the next step.
		No	Go to Step 7.
6	**INSPECT CKP SENSOR SIGNAL** • Verify the PCM terminal 1E signal using the oscilloscope. (See PCM INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Visually inspect the CKP sensor pulse wheel. • If there is any malfunction: — Replace the CKP sensor pulse wheel, then go to Step 12. • If there is no malfunction: — Replace the CKP sensor, then go to Step 12. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to Step 12.
7	**INSPECT VACUUM HOSE CONNECTION** • Inspect the vacuum hose routine. (See INTAKE-AIR SYSTEM VACUUM HOSE ROUTING DIAGRAM [SKYACTIV-D 1.5].) • Are vacuum hoses routines properly?	Yes	Go to the next step.
		No	Reconnect the vacuum hose, then go to Step 12. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8	**INSPECT INTAKE-AIR SYSTEM FOR AIR SUCTION** • Inspect the following parts of the intake air system for looseness, damage, and loss. — Between air cleaner and intake manifold • Is there any leakage?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12.
		No	Go to the next step.
9	**INSPECT HP-EGR CONTROL VALVE CONTROL SYSTEM OPERATION** • Perform the HP-EGR Control Valve Operation Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12. (See HP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**INSPECT LP-EGR CONTROL VALVE CONTROL SYSTEM OPERATION** • Perform the LP-EGR Control Valve Operation Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 12. (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
11	**INSPECT TIMING CHAIN INSTALLATION CONDITION** • Verify the condition of the timing chain assembly (valve timing, looseness, jumping). (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results. Assemble the timing chain using the correct timing, then go to the next step. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
12	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
13	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.